

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I G.Mohammad Shareef / 192311288 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION: G.Mohammad Shareef

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

Answers

1. Define SOAP.

SOAP (Simple Object Access Protocol) is a protocol used for exchanging structured information between applications over a network. It commonly uses XML for message formatting.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based document that describes a web service, including its operations, input/output messages, protocols, and endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules and methods that allows one software application to communicate and interact with another application or service.

4. Define XML Schema.

XML Schema (XSD) is used to define the structure, elements, attributes, data types, and rules of an XML document.

5. What is a Web Service?

A **Web Service** is a software system that allows different applications to communicate and exchange data over a network using standard web technologies.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services using HTTP methods such as GET, POST, PUT, and DELETE.

7. What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted and communicated between devices or applications over a network.

8. Define Endpoint in web services.

An **endpoint** is a specific URL through which a client accesses a web service or API.

9. What is a Service in web applications?

A **service** is a software component that performs a specific function and provides that functionality to other applications through an interface such as an API.

10. What is the difference between SOAP and REST?

SOAP	REST
Protocol	Architectural style
Mainly uses XML	Commonly uses JSON, but can use XML and other formats
More structured and formal	Lightweight and flexible
Uses SOAP-specific standards	Uses standard HTTP methods
Suitable for complex enterprise services	Commonly used for web and mobile APIs

11. What is the role of WSDL in web services?

WSDL describes what a web service does, how to communicate with it, what data it accepts/returns, and where the service is located. It helps clients understand and use SOAP-based services.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight data-interchange format. APIs commonly use JSON to send structured data between clients and servers.

13. What is HTTP and why is it used in web services?

HTTP (Hypertext Transfer Protocol) is a protocol used to transfer data between clients and servers. Web services use HTTP because it provides standard methods such as **GET, POST, PUT, and DELETE** for communication.

14. What is meant by request and response in web communication?

A **request** is a message sent by a client to a server asking for data or an operation.

A **response** is the message returned by the server containing the requested data or the result of the operation.

15. What is the importance of XML in web services?

XML provides a structured, platform-independent format for representing and exchanging data. It is widely used in SOAP messages and can also be used in other web-service technologies.

16. What is the purpose of an API endpoint?

An API endpoint provides a specific access point through which a client can request or send data to a particular resource or service.

17. What is the difference between HTTP and HTTPS?

HTTP	HTTPS
Hypertext Transfer Protocol	Hypertext Transfer Protocol Secure
Data is generally transmitted without encryption	Data is encrypted
Less secure	More secure
Uses HTTP	Uses HTTP with TLS encryption

HTTPS is preferred when sensitive information such as passwords or payment details is transmitted.

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They use resources identified by URLs and standard HTTP methods such as GET, POST, PUT, and DELETE.

19. What is a namespace in XML?

An XML namespace is used to uniquely identify XML elements and avoid naming conflicts when elements from different XML vocabularies are used together.

20. What is the role of a client and server in web services?

The client sends requests to access a service or resource, while the server receives the request, processes it, and sends a response.